MARK SCHEME · TEACHER COPY

10BR · Chapter 2: Data Transmission

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · Oct/Nov 2023 Paper 11, original Q7 [4 marks]

Question	Answer	Marks
1	**One** mark for each correct term, in the correct order: – header – destination address – routers – last	4

Question 2 · Feb/March 2024 Paper 12, original Q7 (a),(b) [8 marks]

Question	Answer	Marks
2(a)(i) i)	If the data is intercepted, it cannot be understood	1
2(a)(ii)	**Four** from: • Symmetric has a shared key… • … to encrypt **and** decrypt • Both the sender and receiver know the key • Asymmetric has different keys // a public key **and** a private key • …public to encrypt the data **and** private to decrypt • …anyone can know the public key but only those intended know the private key	4
2(b)(i)	Any **two** from: e.g. • Destination address/IP • Sender address/IP • Packet number	2
2(b)(ii)	Any **one** from: • Control the route the packet takes • Send each packet towards its destination • Choose more efficient route	1

Question 3 · May/June 2024 Paper 11, original Q4 (a),(b),(c) [7 marks]

Question	Answer	Marks
3(a)	Any **three** from: • A packet is split into **three** different sections • … the header • … the payload • … the trailer	3
3(b)	Router	1
3(c)	Any **three** from: • The network may be spread over a long distance … • … so it is more **reliable** • Bits will be sent/arrive in **sequence** • …so bits less likely to be skewed • Less crosstalk/interference • … so less likely to have errors • The data may not need to be transmitted at a fast speed // data transmission speed of serial is adequate • The cables in the network only use serial transmission	3

Question 4 · May/June 2024 Paper 12, original Q6 [5 marks]

Question	Answer	Marks
4	**One** mark for each correct method.	5

error detection method	statement
parity (check/bit/byte/block)	An odd or even process can be used.
checksum	A value is calculated from the data using an algorithm. This happens before and after the data is transmitted.
echo check	A copy of the data is sent back to the sender by the receiver.
automatic repeat query/request // ARQ	Acknowledgement and timeout are used.

	check digit	A value is appended to data that has been calculated using the data. This value is checked on data entry.

Question 5 · Oct/Nov 2024 Paper 11, original Q4 (a),(b) [11 marks]

Question	Answer	Marks
5(a)	Any **four** from: • The data packet has three sections • It has a packet header … • … that contains data such as the destination address • It has a payload … • … that contains the main data for the email • It has a trailer … • … that contains data such as the error detection system used	**4**
5(b)(i)	Any **four** from: • It sends the data multiple bits at the same time // It uses multiple wires • … so the transmission speed of the data will be fast • Data may not need to travel a long distance • … as the devices are all within a single room • It sends data in both directions at the **same time** • … so users on the network can send data to each other with **no delay**	**4**
5(b)(ii)	Any **two** from: • More interference/crosstalk (due to multiple wires) • **Data** may be skewed (due to multiple bits at a time) // bits may arrive out of order • More chance of data collisions (as data sent in both directions at the same time) • More chance of error in the data	**2**
5(b)(iii)	Any **one** from: • Serial simplex • Serial half-duplex • Serial full-duplex • Parallel simplex • Parallel half-duplex	**1**

Question 6 · May/June 2025 Paper 11, original Q4 (a),(b),(c) [12 marks]

Question	Answer	Marks
6(a)(i)	**Two** from (one mark for serial, one mark for full-duplex): **serial** • Serial is suitable as data may be travelling a long distance (over 5 metres). • Serial is suitable to make sure that bits don't arrive skewed/arrives in order. • Serial is suitable as there is less chance of error. • Serial has sufficient transmission speed as not much data being sent. **full-duplex** • Full-duplex is suitable as data needs to go between customer and kitchen computer // data needs to go in both directions // so notification of errors can be sent back to customer.	**2**
6(a)(ii)	The data may be transmitted faster.	**1**
6(b)(i)	Any **five** from: • The checksum is **calculated** from/using the **data**. • … using an algorithm // by example. • The (checksum) value is transmitted with the data. • The (checksum) value is recalculated **after transmission/by recipient**. • If the **checksum values** do not match, an error is detected // If the **checksum values** match, no error is detected … • … request for data to be resent if there is an error.	**5**
6(b)(ii)	Any **one** from: • Parity byte/block check • Echo check	**1**
6(c)(i)	D	**1**
6(c)(ii)	Any **Two** from: • The packets **arrive** in the wrong order. • Each packet could take a different route. • Some packets may take longer than others to be transmitted. • Some packets may not arrive/be corrupted and need to be requested again.	**2**

Total: 75 marks

Question 7 · May/June 2025 Paper 13, original Q7 (a),(b) [10 marks]

Question	Answer	Marks
7(a)	• It can suffer Interference/crosstalk. • … that could have caused data loss/gained/changed.	2
7(b)	**One** mark for each correct term in the correct place. • odd/even • even/odd (must be opposite of MP1) • bit • echo check • automatic repeat query (ARQ) • timeout • positive/negative or negative/positive A parity check is set to be **odd** or **even**. A parity **bit** is added to each byte before transmission to make it match the set parity. An **echo check** involves comparing the data sent to the data received back from the receiving device. An **automatic repeat query (ARQ)** uses acknowledgement and **timeout**. The acknowledgement system can be **positive** or **negative**.	8

Question 8 · Oct/Nov 2025 Paper 13, original Q3 (a),(b) [14 marks]

Question	Answer	Marks

8(a)	**One** mark for each correct section **One** mark for each correct example of what it stores First section • Header • Originators/senders (IP) address • Destination/receivers (IP) address • Packet number Second section • Payload • The data itself Third section • Trailer • End of packet notification • Error checking data	**6**
8(b)(i)	**Four** from: • Data is sent one bit at a time • … down a single wire • Data can be sent in both directions … • … at the same time	**4**

8(b)(ii)	Any **Four** from: • Data is sent multiple <u>bits</u> at the **same time** • … down multiple wires • Data can be sent in both directions … • … not at the same time	**4**

Question 9 · May/June 2026 Paper 11, original Q4 (b) [4 marks]

Question	Answer	Marks	Guidance
9 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
9(a)	**Four** from: • (Serial) transmits the data one bit at a time • … down a single wire/channel/line • (Half-duplex) transmits the data (between the computer and the mouse) in both directions/ways … • … but only one direction/way at a time	**4**	MP1 – BOD bit by bit MP2 –cable/path is TV